RP2040 Microcontroller Datasheet - DATASHEET

Manufacturer: Raspberry Pi
Part Number: RP2040
Category: Microcontroller / Dual-Core ARM MCU
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Overview

RP2040 is a low-cost, high-performance microcontroller with flexible digital interfaces. Dual ARM Cortex-M0+ cores at up to 133 MHz. 264 KB on-chip SRAM in six independent banks. External Quad-SPI Flash support with execute-in-place (XIP). Programmable I/O (PIO) subsystem: 2 PIO blocks with 4 state machines each for custom peripheral implementations without CPU overhead.

PIO Subsystem

Programmable I/O (PIO) is a versatile hardware interface. Each PIO block has 4 state machines that execute sequential programs from a shared instruction memory of 32 instructions. The PIO instruction set includes: JMP, WAIT, IN, OUT, PUSH, PULL, MOV, IRQ, SET. State machines can shift data in/out at up to system clock speed (133 MHz). Ideal for implementing WS2812B LED drivers, VGA video output, SDIO, I2S, and other protocols.

Key Specifications Summary

cores: 2xCortexM0+. freq: 133MHz. sram: 264KB. pio: 8SM. gpio: 30. adc: 4ch_12bit. usb: device.